

DAY-2

Tasks

1.

List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware issue	Typical symptom
1. RAM failure	PC may not boot, may show a blank screen, or produce beep codes
2. Hard disk/SSD failure	Slow booting, operating system not loading, or “No boot device” error
3. Power supply failure	PC does not turn on, randomly shuts down, or restarts
4. Overheating	PC becomes very hot, freezes, slows down, or shuts down suddenly
5. Graphics card failure	No display, screen artifacts, flickering, or crashes during graphics use

2.

Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step 1: Check the wall socket

- Plug another working device into the socket.
- Check whether the socket is providing power.
- If there is no power, try another socket.

Step 2: Inspect the charger

- Check the charger, cable, and connector for cuts, burns, loose parts, or bent pins.
- If the cable is damaged, the charger may not provide power properly.

Step 3: Check the charger's output voltage

- Set the multimeter to DC voltage.
- Place the black probe on the outer/negative part of the charger connector.
- Place the red probe on the positive pin.
- Compare the reading with the voltage printed on the charger label.

Step 4: Check the charging port

- Inspect the laptop's charging port for dust, damage, or looseness.
- Gently check whether the connector moves excessively when inserted.

Step 5: Check the battery

- If the battery is removable, remove it and inspect it for swelling or physical damage.
- A damaged or completely failed battery can prevent normal charging.

Step 6: Check connections

- Make sure the charger connector is fully inserted.
- If possible, test with a known-good compatible charger.

Step 7: Identify the faulty component

If the wall socket and charger are working but the laptop still does not charge, the problem may be with the charging port, battery, charging circuit, or motherboard.

3.

Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Procedure

1. Set the multimeter to DC voltage (V $\equiv$).
2. Connect the charger to a power outlet.
3. Insert a USB breakout/test cable or suitable USB test leads.
4. Touch the black probe to the USB ground (GND).
5. Touch the red probe to the USB 5V/VBUS pin.
6. Read the voltage on the multimeter.

Sample result

Measured voltage: 5.02 V DC

Expected range: 4.75 V – 5.25 V

Result: 5.02 V is within the expected range, so the charger is providing a normal USB voltage.

4.

A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Procedure

1. Shut down the PC and disconnect it from the wall.
2. Disconnect the ATX power connector from the motherboard.
3. Set the multimeter to DC voltage (V $\overline{=}$).
4. Reconnect the PSU to AC power only if your test setup requires the PSU to be powered.
5. Identify the yellow wire — this is the +12V rail.
6. Identify a black wire — this is ground.
7. Place the black multimeter probe on the black wire.
8. Place the red probe on the yellow wire.
9. Read the voltage displayed on the multimeter.